

Comprehensive Questions and Worked Solutions

Lecture 1: Classical Linear Regression

Lecture 2: Univariate Time Series (1)

Lecture 3: Univariate Time Series (2)

Based on the supplied ECON 3350/7350 lecture slides

Scope and notation

This question set covers the substantive teaching material in the three supplied lectures. It includes conceptual questions, derivations, numerical calculations, interpretation of the lecture examples, and integrated applications. Course administration, historical assessment dates, and staff contact details are excluded.

Some derivations expand on results stated briefly in the slides. Numerical examples are constructed for practice unless explicitly identified as lecture examples.

Throughout the time-series questions, innovations have mean zero and constant variance σ_ε^2 . For conditional-expectation forecasting, we additionally assume that future innovations have conditional mean zero given past information. Serial uncorrelatedness alone is not sufficient for that stronger property.

Corrections and qualifications to the slides

The solutions use the following corrected statements:

- The first partial autocorrelation is $\phi_{11} = \rho_1$, not γ_1 .
- The Ljung–Box statistic has denominator $T - k$, not $T + k$.
- Changing dollars to cents multiplies covariances by 100²; correlations remain unchanged.
- MLE does not universally guarantee the lowest finite-sample MSE. Its efficiency claims require suitable assumptions and are usually asymptotic.
- Minimizing an ordinary residual sum of squares estimates the dynamic coefficients; the innovation variance is then estimated from the residuals.
- Lag-polynomial representations without an intercept are written for a zero-mean or demeaned series.

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1 Lecture 1: Classical Linear Regression

Question 1: Model, parameters, and matrix dimensions

Write the multiple linear regression model in scalar and matrix form. Define its components and their dimensions. Does linearity exclude squared regressors?

Worked solution

The scalar model is

$$y_i = \beta_1 + \beta_2 x_{2,i} + \cdots + \beta_k x_{k,i} + \varepsilon_i, \quad i = 1, \dots, n.$$

Here y_i is the dependent variable, the x variables are regressors, the β coefficients are unknown parameters, and ε_i represents omitted influences.

Stacking observations gives

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{y}, \boldsymbol{\varepsilon} \in \mathbb{R}^{n \times 1}, \quad \mathbf{X} \in \mathbb{R}^{n \times k}, \quad \boldsymbol{\beta} \in \mathbb{R}^{k \times 1}.$$

The first column of $\mathbf{X}$ is a vector of ones when an intercept is included.

Linearity means linearity in the parameters. For example,

$$y_i = \beta_1 + \beta_2 x_i + \beta_3 x_i^2 + \varepsilon_i$$

is a linear regression model because the unknown coefficients enter linearly.

Question 2: Classical regression assumptions

State the five assumptions discussed in Lecture 1. Explain the role of each assumption.

Worked solution

(a) **Linearity:**

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i.$$

The specified conditional relationship is linear in parameters.

(b) **Exogeneity:**

$$\mathbb{E}(\varepsilon_i \mid \mathbf{x}_i) = 0.$$

The regressors do not systematically predict the error's mean.

(c) **Homoskedasticity:**

$$\text{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2.$$

Conditional error variance is constant across observations.

(d) **No cross-observation error correlation:**

$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid \mathbf{x}_i, \mathbf{x}_j) = 0, \quad i \neq j.$$

The slides associate this with random sampling. Independent random sampling is stronger than merely having uncorrelated errors.

(e) **No perfect multicollinearity:** The regressor matrix has full column rank:

$$\text{rank}(\mathbf{X}) = k.$$

Therefore $\mathbf{X}'\mathbf{X}$ is invertible.

For matrix-based finite-sample OLS results, we use the corresponding conditions

$$\mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n.$$

Question 3: Deriving the OLS estimator

Derive OLS by minimizing the sum of squared residuals. Which assumption guarantees a unique solution?

Worked solution

Define

$$S(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b}).$$

Expanding,

$$S(\mathbf{b}) = \mathbf{y}'\mathbf{y} - 2\mathbf{b}'\mathbf{X}'\mathbf{y} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b}.$$

Differentiate:

$$\frac{\partial S(\mathbf{b})}{\partial \mathbf{b}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\mathbf{b}.$$

Setting the derivative to zero yields

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}.$$

Hence

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Full column rank, or no perfect multicollinearity, ensures invertibility and a unique minimizer.

The residual vector satisfies

$$\mathbf{X}'\hat{\boldsymbol{\varepsilon}} = \mathbf{0}.$$

If an intercept is included, the residuals sum to zero.

Question 4: Unbiasedness, variance, and BLUE

Show that OLS is unbiased under the classical assumptions and derive its conditional covariance matrix. Explain BLUE.

Worked solution

Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}.$$

Therefore,

$$\mathbb{E}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \boldsymbol{\beta}.$$

Using $\text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$,

$$\begin{aligned} \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2 \mathbf{I}_n)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \end{aligned}$$

BLUE means *Best Linear Unbiased Estimator*. Among estimators linear in $\mathbf{y}$ and conditionally unbiased, OLS has the smallest covariance matrix in the Gauss–Markov sense.

It does not mean OLS dominates all biased or nonlinear estimators under every loss function.

Question 5: MSE and the bias–variance tradeoff

Derive the scalar MSE decomposition. An unbiased estimator has variance 4. A second estimator has bias 1 and variance 2. Which has lower MSE? State the vector version.

Worked solution

Write

$$\hat{\beta} - \beta = (\hat{\beta} - \mathbb{E}\hat{\beta}) + (\mathbb{E}\hat{\beta} - \beta).$$

Squaring and taking expectations eliminates the cross term:

$$\text{MSE}(\hat{\beta}) = \text{Var}(\hat{\beta}) + \text{Bias}(\hat{\beta})^2.$$

For the unbiased estimator,

$$\text{MSE}_1 = 4.$$

For the biased estimator,

$$\text{MSE}_2 = 2 + 1^2 = 3.$$

The biased estimator has lower MSE because its variance reduction more than compensates for its squared bias.

For a vector estimator under total squared-error loss,

$$\text{MSE}(\hat{\beta}) = \text{tr Var}(\hat{\beta}) + \|\mathbb{E}(\hat{\beta}) - \beta\|^2.$$

Question 6: Unbiasedness, consistency, and MLE

Distinguish unbiasedness from consistency. Give an example showing that neither implies the other. Is MLE always the lowest-MSE estimator?

Worked solution

Unbiasedness is a finite-sample statement:

$$\mathbb{E}(\hat{\theta}_n) = \theta.$$

Consistency is a probability-limit statement:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty.$$

Suppose $X_1, \dots, X_n$ are i.i.d. with mean θ and positive finite variance.

The estimator $\hat{\theta}_n = X_1$ is unbiased, but its distribution does not become more concentrated as n grows. It is not consistent.

The estimator

$$\tilde{\theta}_n = \bar{X}_n + \frac{1}{n}$$

is biased by $1/n$, but it is consistent because $\bar{X}_n \xrightarrow{p} \theta$ and $1/n \rightarrow 0$.

MLE is not universally optimal in finite-sample MSE. Under correct specification and regularity conditions, it is generally consistent and asymptotically efficient within the relevant regular class.

With conditionally normal, homoskedastic regression errors, maximizing the likelihood over β is equivalent to minimizing squared residuals, so the MLE of β is OLS.

Question 7: Wages, conditional expectations, and causality

Consider

$$\text{wage}_i = \beta_1 + \beta_2 \text{YOE}_i + \beta_3 \text{YOS}_i + \varepsilon_i.$$

Derive the conditional mean and interpret β_3 . Why does regression alone not establish causality?

Worked solution

Under exogeneity,

$$\mathbb{E}(\text{wage}_i \mid \text{YOE}_i, \text{YOS}_i) = \beta_1 + \beta_2 \text{YOE}_i + \beta_3 \text{YOS}_i.$$

Therefore,

$$\frac{\partial \mathbb{E}(\text{wage}_i \mid \text{YOE}_i, \text{YOS}_i)}{\partial \text{YOS}_i} = \beta_3.$$

It measures the conditional association of one additional year of schooling with wages, holding experience fixed.

A causal interpretation needs an identification argument. For example, unobserved ability may influence both schooling and wages, making schooling correlated with ε_i .

If schooling changes experience, then a total causal effect may also include the effect transmitted through experience. A coefficient holding experience fixed does not automatically measure that total effect.

Reversing the dependent and explanatory variables produces a different conditional relationship; neither regression direction establishes causality by itself.

Question 8: Individual and joint hypothesis tests

For the wage model, formulate tests of:

- (a) No schooling association, conditional on experience.
- (b) No association with either schooling or experience.
- (c) Explain why failure to reject is not acceptance.

Worked solution

The individual hypotheses are

$$H_0 : \beta_3 = 0, \quad H_1 : \beta_3 \neq 0.$$

A usual statistic is

$$t = \frac{\hat{\beta}_3}{\text{se}(\hat{\beta}_3)}.$$

Its exact finite-sample t distribution requires additional normal-error assumptions; asymptotic inference uses suitable large-sample conditions.

The joint hypotheses are

$$H_0 : \beta_2 = \beta_3 = 0,$$

against the alternative that at least one restriction is false. They are tested jointly using an F - or Wald test.

For the classical F test with $r = 2$ restrictions,

$$F = \frac{(SSR_R - SSR_U)/r}{SSR_U/(n - k)},$$

where R and U denote restricted and unrestricted models.

Failing to reject means that the evidence is insufficient at the chosen significance level. It does not prove the null. Small samples, noisy outcomes, or imprecise regressors can produce low power.

Question 9: Regressor selection and time-series limitations

Explain the risks of omitting relevant regressors, including irrelevant regressors, selecting only significant regressors, and maximizing R^2 . Why are classical assumptions often problematic in macroeconomics and finance?

Worked solution

Omitting a relevant variable generally creates bias if that variable is correlated with an included regressor. For example, if the true model includes z_i ,

$$y_i = \beta_0 + \beta_1 x_i + \delta z_i + u_i,$$

then, under suitable conditions, omitting z_i gives

$$\text{plim } \hat{\beta}_1 = \beta_1 + \delta \frac{\text{Cov}(x_i, z_i)}{\text{Var}(x_i)}.$$

An irrelevant variable need not bias OLS under exogeneity, but can reduce precision and consume degrees of freedom.

Insignificance alone is not a sufficient reason to remove a variable: low power and multicollinearity can hide an economically important relationship.

Raw R^2 cannot decrease when additional regressors are included. It therefore rewards complexity without penalizing it.

A sensible approach combines theory, multiple defensible specifications, adjusted fit measures, AIC/BIC, and transparent reporting.

Time-series observations may exhibit serial correlation, changing volatility, and feedback between regressors and errors. These threaten the classical exogeneity and error-covariance assumptions and motivate dynamic models.

2 Lecture 2: Univariate Time-Series Foundations

Question 10: Stochastic processes and realizations

Define a stochastic process and a realization. What is the distinction between an ensemble mean and a time average?

Worked solution

A stochastic process $\{y_t\}$ is a collection of random variables indexed by time. A realization is one observed path,

$$y_1, y_2, \dots, y_T,$$

generated by its joint probability distribution.

The ensemble mean

$$\mu_t = \mathbb{E}(y_t)$$

averages the possible outcomes at a fixed date across hypothetical realizations.

The time average

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t$$

averages observations across dates in one realization.

They are conceptually different. Under appropriate stationarity and ergodicity conditions, the time average can consistently estimate the population mean.

Question 11: Moments and covariance stationarity

Define the first and second moments of a time series. State covariance stationarity and distinguish it from having a constant mean alone.

Worked solution

The relevant moments are

$$\mu_t = \mathbb{E}(y_t),$$

$$\text{Var}(y_t) = \mathbb{E}[(y_t - \mu_t)^2],$$

and

$$\text{Cov}(y_t, y_{t-k}) = \mathbb{E}[(y_t - \mu_t)(y_{t-k} - \mu_{t-k})].$$

Covariance stationarity requires finite second moments and

$$\mathbb{E}(y_t) = \mu, \quad \text{Var}(y_t) = \gamma_0, \quad \text{Cov}(y_t, y_{t-k}) = \gamma_k$$

for all dates t .

Thus covariance depends on the lag k , not the calendar date. A constant mean alone is insufficient: a process with variance increasing over time is not covariance stationary.

Stationarity does not mean that observations are constant or independent. Persistent fluctuations around a stable mean are compatible with stationarity.

Question 12: AR(1) forecasts and the long-run mean

For

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t,$$

derive the h -step conditional-mean forecast and its limit when $|a_1| < 1$.

Worked solution

Let $\mathcal{F}_T$ denote information available at time T , and assume

$$\mathbb{E}(\varepsilon_{T+j} \mid \mathcal{F}_T) = 0, \quad j \geq 1.$$

The first forecasts are

$$\hat{y}_{T+1|T} = a_0 + a_1 y_T,$$

$$\hat{y}_{T+2|T} = a_0 + a_1(a_0 + a_1 y_T) = a_0(1 + a_1) + a_1^2 y_T.$$

Continuing,

$$\hat{y}_{T+h|T} = a_0 \sum_{j=0}^{h-1} a_1^j + a_1^h y_T.$$

When $a_1 \neq 1$,

$$\hat{y}_{T+h|T} = a_0 \frac{1 - a_1^h}{1 - a_1} + a_1^h y_T.$$

Define

$$\mu = \frac{a_0}{1 - a_1}.$$

Then

$$\hat{y}_{T+h|T} = \mu + a_1^h (y_T - \mu).$$

If $|a_1| < 1$, the effect of today's deviation vanishes:

$$\hat{y}_{T+h|T} \rightarrow \mu.$$

Positive a_1 gives monotone mean reversion in the forecast; negative a_1 gives alternating convergence.

Question 13: AR(1) variance, autocorrelation, and boundaries

Derive the stationary AR(1) variance and ACF. Explain what changes at $a_1 = 1$, $a_1 = -1$, and $|a_1| > 1$ under the usual causal specification.

Worked solution

Demean the process:

$$x_t = y_t - \mu = a_1 x_{t-1} + \varepsilon_t.$$

Because the innovation is uncorrelated with past x ,

$$\text{Var}(x_t) = a_1^2 \text{Var}(x_{t-1}) + \sigma_\varepsilon^2.$$

Under stationarity,

$$\gamma_0 = a_1^2 \gamma_0 + \sigma_\varepsilon^2,$$

so

$$\gamma_0 = \frac{\sigma_\varepsilon^2}{1 - a_1^2}.$$

For $k \geq 1$,

$$\gamma_k = a_1 \gamma_{k-1},$$

which implies

$$\rho_k = a_1^k.$$

At $a_1 = 1$, the model is a random walk, possibly with drift:

$$y_{T+h} = y_T + h a_0 + \sum_{j=1}^h \varepsilon_{T+j}.$$

Its forecast-error variance is $h \sigma_\varepsilon^2$.

At $a_1 = -1$, shocks also fail to decay in magnitude, and forecast paths typically oscillate. For $|a_1| > 1$, the usual forward causal process is explosive. The stationary causal AR(1) condition is $|a_1| < 1$.

Question 14: AR, MA, ARMA, and white noise

Write $AR(p)$, $MA(q)$, and $ARMA(p, q)$ models. Define white noise and explain the roles of p and q .

Worked solution

The models are

$$AR(p) : \quad y_t = a_0 + \sum_{j=1}^p a_j y_{t-j} + \varepsilon_t,$$

$$MA(q) : \quad y_t = a_0 + \varepsilon_t + \sum_{j=1}^q \beta_j \varepsilon_{t-j},$$

$$ARMA(p, q) : \quad y_t = a_0 + \sum_{j=1}^p a_j y_{t-j} + \varepsilon_t + \sum_{j=1}^q \beta_j \varepsilon_{t-j}.$$

Here p counts lagged dependent variables and q counts lagged innovations. Weak white noise satisfies

$$\mathbb{E}(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = \sigma_\varepsilon^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s).$$

White noise need not be independent or normally distributed.

An MA model is a weighted sum of innovations. It is not simply a moving average of observed y values.

Question 15: Special cases, means, and infinite representations

For $ARMA(1,1)$, identify the AR and MA special cases. Derive its mean. State the usual stationarity and invertibility conditions, and explain their different roles.

Worked solution

Consider

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t + \beta_1 \varepsilon_{t-1}.$$

If $\beta_1 = 0$, it is $AR(1)$. If $a_1 = 0$, it is $MA(1)$. If both vanish, $y_t = a_0 + \varepsilon_t$.

Taking expectations,

$$\mu = a_0 + a_1 \mu,$$

so

$$\mu = \frac{a_0}{1 - a_1}.$$

More generally, for stationary $ARMA(p, q)$,

$$\mu = \frac{a_0}{1 - \sum_{j=1}^p a_j}.$$

For a minimal $ARMA(1,1)$ specification:

- $|a_1| < 1$ gives a causal stationary $MA(\infty)$ representation.
- $|\beta_1| < 1$ gives an invertible $AR(\infty)$ representation.

Stationarity concerns stable moments and decay of past shocks. Invertibility concerns recovering innovations from present and past observations.

These are distinct: a finite MA process is stationary even when its usual invertibility condition fails.

Question 16: ACF, sample ACF, and measurement units

Define the ACF and give a common sample estimator. Explain symmetry and the effect of changing units.

Worked solution

For a stationary process,

$$\rho_k = \frac{\gamma_k}{\gamma_0}, \quad \gamma_k = \text{Cov}(y_t, y_{t-k}).$$

The ACF is the sequence of ρ_k values plotted against lag.

A common sample estimator is

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}.$$

Plotting r_k gives the sample ACF or correlogram.

Covariance symmetry and stationarity imply

$$\gamma_{-k} = \gamma_k, \quad \rho_{-k} = \rho_k.$$

If $z_t = cy_t + d$ with $c \neq 0$, then

$$\gamma_k^z = c^2 \gamma_k^y, \quad \rho_k^z = \rho_k^y.$$

Thus cents have larger numerical covariances than dollars, but exactly the same autocorrelations.

Also,

$$\rho_0 = 1, \quad -1 \leq \rho_k \leq 1.$$

Question 17: PACF and the lag-two formula

Define the PACF and derive

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

Calculate it for $\rho_1 = 0.6$ and $\rho_2 = 0.2$.

Worked solution

The lag- k PACF measures the linear association between y_t and y_{t-k} after removing the linear effects of the intervening observations.

Equivalently, for a stationary demeaned series, ϕ_{kk} is the last coefficient in its population linear projection on the previous k observations.

At lag two, the Yule–Walker equations are

$$\rho_1 = \phi_{21} + \phi_{22}\rho_1,$$

$$\rho_2 = \phi_{21}\rho_1 + \phi_{22}.$$

The first equation gives

$$\phi_{21} = \rho_1(1 - \phi_{22}).$$

Substitute into the second:

$$\rho_2 = \rho_1^2(1 - \phi_{22}) + \phi_{22}.$$

Therefore,

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

Numerically,

$$\phi_{22} = \frac{0.2 - 0.36}{1 - 0.36} = -0.25.$$

At lag one,

$$\phi_{11} = \rho_1.$$

Question 18: General PACF recursion and AR order

State the recursion for obtaining PACF values from the ACF. Why does the PACF cut off for an $AR(p)$?

Worked solution

Starting with $\phi_{11} = \rho_1$, calculate

$$\phi_{kk} = \frac{\rho_k - \sum_{j=1}^{k-1} \phi_{k-1,j} \rho_{k-j}}{1 - \sum_{j=1}^{k-1} \phi_{k-1,j} \rho_j}.$$

Then update the other coefficients:

$$\phi_{k,j} = \phi_{k-1,j} - \phi_{kk} \phi_{k-1,k-j}, \quad j = 1, \dots, k-1.$$

For a causal $AR(p)$,

$$y_t - \mu = \sum_{j=1}^p a_j (y_{t-j} - \mu) + \varepsilon_t.$$

Once the previous p observations are included, more distant lags provide no additional linear predictive information, because ε_t is uncorrelated with the past.

Hence

$$\phi_{pp} = a_p, \quad \phi_{kk} = 0 \quad \text{for } k > p.$$

The population PACF cuts off after lag p . Sample PACFs fluctuate around these population values.

Question 19: Lecture $AR(2)$ example

For the lecture model

$$y_t = 0.7y_{t-1} - 0.49y_{t-2} + \varepsilon_t,$$

calculate ρ_1 , ρ_2 , ϕ_{11} , and ϕ_{22} . Describe its ACF and PACF.

Worked solution

For $AR(2)$, the first Yule–Walker equation is

$$\rho_1 = a_1 + a_2 \rho_1.$$

Therefore,

$$\rho_1 = \frac{a_1}{1 - a_2} = \frac{0.7}{1.49} \approx 0.4698.$$

Next,

$$\rho_2 = a_1\rho_1 + a_2 = 0.7(0.4698) - 0.49 \approx -0.1611.$$

The PACF values are

$$\phi_{11} = \rho_1 \approx 0.4698,$$

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = -0.49.$$

All later population PACF values are zero.

The ACF follows

$$\rho_k = 0.7\rho_{k-1} - 0.49\rho_{k-2}.$$

The dynamic roots solve

$$r^2 - 0.7r + 0.49 = 0$$

and have modulus $0.7 < 1$. Consequently, the ACF displays damped oscillation, while the PACF cuts off after lag two.

Question 20: Lecture MA(2) example

For

$$y_t = \varepsilon_t + 0.7\varepsilon_{t-1} - 0.49\varepsilon_{t-2},$$

derive its variance, first two autocovariances, and ACF. How does its PACF differ from the AR(2) example?

Worked solution

For MA(2),

$$y_t = \varepsilon_t + b_1\varepsilon_{t-1} + b_2\varepsilon_{t-2},$$

only matching innovation dates contribute to covariance. Thus

$$\gamma_0 = (1 + b_1^2 + b_2^2)\sigma_\varepsilon^2,$$

$$\gamma_1 = (b_1 + b_1b_2)\sigma_\varepsilon^2, \quad \gamma_2 = b_2\sigma_\varepsilon^2.$$

At lags above two, no innovations overlap, so $\gamma_k = 0$.

For $b_1 = 0.7$ and $b_2 = -0.49$,

$$\gamma_0 = 1.7301\sigma_\varepsilon^2, \quad \gamma_1 = 0.357\sigma_\varepsilon^2, \quad \gamma_2 = -0.49\sigma_\varepsilon^2.$$

Hence

$$\rho_1 = \frac{0.357}{1.7301} \approx 0.2063,$$

$$\rho_2 = \frac{-0.49}{1.7301} \approx -0.2832,$$

and

$$\rho_k = 0 \quad (k > 2).$$

The MA(2) ACF cuts off after lag two, whereas its PACF generally tails off. This reverses the cutoff pattern of an AR(2).

Question 21: GDP transformations and identification patterns

Why does the lecture examine first differences of log real GDP per capita? Interpret the transformation and summarize the ACF/PACF rules for AR and MA models.

Worked solution

GDP levels often trend. Taking logs expresses changes proportionally, while first differencing may remove a stochastic trend:

$$g_t = \Delta \log Y_t = \log Y_t - \log Y_{t-1}.$$

For small changes,

$$100g_t \approx 100 \frac{Y_t - Y_{t-1}}{Y_{t-1}},$$

the percentage growth rate.

The exact percentage growth rate is

$$100(e^{g_t} - 1).$$

Transformation does not guarantee stationarity; the transformed series still needs inspection. For quarterly data, $400g_t$ is an approximate annualized percentage growth rate. The standard population identification patterns are:

Model	ACF	PACF
AR(1)	Tails off	Cuts off after 1
AR(2)	Tails off	Cuts off after 2
AR(p)	Tails off	Cuts off after p
MA(1)	Cuts off after 1	Tails off
MA(q)	Cuts off after q	Tails off
ARMA(p, q)	Usually tails off	Usually tails off

These are identification guides, not exact rules for noisy sample correlograms. Special parameter cancellations may produce a lower-order model.

3 Lecture 3: Identification, Estimation, and Diagnostics**Question 22: The dividend-yield identification example**

In the lecture's Coles Myer dividend-yield example, the first sample autocorrelation is 0.818, the ACF decays, and the PACF has a dominant first-lag spike. What model is suggested? What can a time plot establish?

Worked solution

The combination of a tailing ACF and a PACF concentrated at lag one suggests AR(1). The large positive first autocorrelation indicates persistence: a high dividend yield tends to be followed by another high value.

The time plot is useful for inspecting trends, level shifts, volatility changes, and unusual observations. The absence of an obvious sustained trend makes a stationary model plausible, but a plot alone cannot prove stationarity.

The lecture's raw-series Ljung–Box tests reject zero autocorrelation, supporting the need for a dynamic model.

Nearby models should still be estimated because sample ACF and PACF patterns are imperfect.

Question 23: Ljung–Box hypotheses and calculation

State the Ljung–Box test and calculate it for

$$T = 100, \quad K = 2, \quad r_1 = 0.20, \quad r_2 = 0.10.$$

Use the 5% critical value $\chi_{2,0.95}^2 = 5.991$.

Worked solution

The hypotheses are

$$H_0 : \rho_1 = \cdots = \rho_K = 0,$$

H_1 : at least one of these autocorrelations is nonzero.

The statistic is

$$Q_K = T(T+2) \sum_{k=1}^K \frac{r_k^2}{T-k}.$$

Under the usual null conditions for an unmodeled white-noise series, its reference distribution is approximately χ_K^2 .

Here,

$$Q_2 = 100(102) \left(\frac{0.04}{99} + \frac{0.01}{98} \right) \approx 5.162.$$

Since $5.162 < 5.991$, we fail to reject at 5%.

The test jointly examines the first K correlations. Failure to reject does not prove independence, normality, or absence of dependence at every other lag.

Question 24: Why AR models permit OLS but MA terms complicate it

Explain the estimation difference between a pure AR model and an ARMA model with MA terms. Is OLS in an AR model necessarily finite-sample unbiased?

Worked solution

In a pure AR model,

$$y_t = a_0 + \sum_{j=1}^p a_j y_{t-j} + \varepsilon_t,$$

the regressors are observed lagged outcomes. The equation is linear in its coefficients, so OLS can be computed directly.

With MA terms,

$$y_t = a_0 + \sum_{j=1}^p a_j y_{t-j} + \varepsilon_t + \sum_{j=1}^q \beta_j \varepsilon_{t-j},$$

the lagged innovations are unobserved. Residuals must be constructed recursively using trial parameter values, making the estimation problem nonlinear.

Although the current innovation may be uncorrelated with past outcomes, it influences

future outcomes that appear elsewhere in the sample's regressor matrix. Strict exogeneity of the entire matrix is therefore generally absent.
 OLS for a stable AR model can be consistent under suitable conditions while still having finite-sample bias.

Question 25: NLLS, likelihood, and Gaussian estimation

Define NLLS and MLE for ARMA models. Explain how the innovation variance is estimated and why Gaussian conditional MLE is related to least squares.

Worked solution

Let η collect the intercept, AR coefficients, and MA coefficients. For trial values of η , compute residuals recursively:

$$e_t(\eta) = y_t - a_0 - \sum_{j=1}^p a_j y_{t-j} - \sum_{j=1}^q \beta_j e_{t-j}(\eta).$$

Conditional nonlinear least squares chooses

$$\hat{\eta} = \arg \min_{\eta} \sum_t e_t(\eta)^2,$$

using an initialization for presample residuals.

A residual-based innovation variance estimate is then

$$\hat{\sigma}_{\varepsilon}^2 = \frac{1}{T_{\text{eff}}} \sum_t e_t(\hat{\eta})^2,$$

with possible degrees-of-freedom adjustments.

A likelihood is the joint density of the observed sample, viewed as a function of unknown parameters.

Under conditionally independent Gaussian innovations, the conditional log-likelihood has the form

$$\ell(\eta, \sigma_{\varepsilon}^2) = -\frac{T_{\text{eff}}}{2} \log(2\pi) - \frac{T_{\text{eff}}}{2} \log \sigma_{\varepsilon}^2 - \frac{\sum_t e_t(\eta)^2}{2\sigma_{\varepsilon}^2}.$$

Concentrating out σ_{ε}^2 links maximization to minimizing the residual sum of squares.

Exact likelihood also accounts for the initial observations' distribution, so it can differ from conditional estimation.

Question 26: AIC, BIC, and the fit–complexity tradeoff

Write general likelihood-based AIC and BIC. Relate them to the lecture's residual-variance formulas. What conditions make comparisons meaningful?

Worked solution

Let k be the number of estimated parameters counted under a consistent convention. General formulas are

$$AIC = -2\hat{\ell} + 2k, \quad BIC = -2\hat{\ell} + k \log T.$$

Lower values are preferred.

The likelihood rewards fit, while the parameter penalty discourages unnecessary complexity.

For $T > e^2$, BIC's penalty per parameter exceeds AIC's.
The lecture gives the simplified, per-observation forms

$$AIC = \log \hat{\sigma}_\varepsilon^2 + \frac{2(p+q+1)}{T},$$

$$BIC = \log \hat{\sigma}_\varepsilon^2 + \frac{(p+q+1) \log T}{T}.$$

These omit constants common to the compared models and use a particular parameter-count convention.

Software may include the innovation variance in k and retain likelihood constants, so reported levels can differ from these simplified formulas without changing rankings.

Compare models using the same dependent variable, estimation sample, and compatible likelihood conventions.

Question 27: Reading the lecture estimation output

The lecture reports:

	$AR(1)$	$MA(1)$
AIC	2.094	2.554
BIC	2.183	2.643

The AR coefficient is approximately 0.8502. Interpret the comparison and explain why a reported constant may not equal the recursion intercept.

Worked solution

Both criteria prefer AR(1) because its values are lower. This agrees with the ACF/PACF identification.

The coefficient 0.8502 is positive and below one in magnitude, suggesting persistent but mean-reverting dynamics.

Some software estimates the mean-adjusted form

$$y_t - \mu = a_1(y_{t-1} - \mu) + \varepsilon_t.$$

Expanding,

$$y_t = (1 - a_1)\mu + a_1y_{t-1} + \varepsilon_t.$$

Thus the recursion intercept is

$$a_0 = (1 - a_1)\mu.$$

If the displayed constant 3.9163 represents μ , then

$$a_0 = (1 - 0.8502)(3.9163) \approx 0.5866.$$

One must check the software parameterization before treating a reported constant as a_0 .

Question 28: Residual diagnostics and adjusted degrees of freedom

Why inspect residuals after model selection? State the residual Ljung–Box degrees-of-freedom adjustment and interpret the lecture's residual comparison.

Worked solution

A satisfactory ARMA model should absorb the serial dependence in the conditional mean. Its residuals should therefore have no systematic autocorrelation pattern.

For residuals from ARMA(p, q), the lecture uses

$$Q_K \stackrel{a}{\sim} \chi^2_{K-p-q},$$

with $K > p + q$.

In the lecture's lag-24 results:

- AR(1): $Q_{24} \approx 24.361$ and $p \approx 0.384$. There is insufficient evidence of remaining autocorrelation.
- MA(1): $Q_{24} \approx 97.980$ with a near-zero p -value. Substantial dependence remains.

The residual evidence further favors AR(1).

Passing this test addresses the selected linear autocorrelations. It does not establish normality, independence, or constant conditional volatility.

4 Lag Operators and Infinite Representations

Question 29: Lag operators, intercepts, and polynomial roots

Define the lag operator and write ARMA(p, q) using lag polynomials. Explain the root conditions and why an intercept must be handled explicitly.

Worked solution

The lag operator satisfies

$$\mathbf{L}y_t = y_{t-1}, \quad \mathbf{L}^j y_t = y_{t-j}.$$

Define

$$a(\mathbf{L}) = 1 - a_1\mathbf{L} - \cdots - a_p\mathbf{L}^p,$$

$$\beta(\mathbf{L}) = 1 + \beta_1\mathbf{L} + \cdots + \beta_q\mathbf{L}^q.$$

Then

$$a(\mathbf{L})y_t = a_0 + \beta(\mathbf{L})\varepsilon_t.$$

If

$$\mu = \frac{a_0}{1 - \sum_{j=1}^p a_j}, \quad x_t = y_t - \mu,$$

the centered representation is

$$a(\mathbf{L})x_t = \beta(\mathbf{L})\varepsilon_t.$$

For a minimal causal stationary ARMA model, the zeros of $a(z)$ lie outside the unit circle. For invertibility, the zeros of $\beta(z)$ lie outside the unit circle.

These polynomial roots are the reciprocals of the dynamic roots sometimes displayed inside the unit circle. The convention must be checked when interpreting output.

Question 30: MA-infinity and AR-infinity representations

Derive the two infinite representations of a centered ARMA process. Explain what each representation is useful for.

Worked solution

Start with

$$a(L)x_t = \beta(L)\varepsilon_t.$$

Under causal stationarity, invert the AR polynomial:

$$x_t = \theta(L)\varepsilon_t, \quad \theta(L) = \frac{\beta(L)}{a(L)}.$$

Writing

$$\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots$$

gives

$$x_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots$$

This MA(∞) representation exposes the effect of past shocks and is central to forecast variances and IRFs.

Under invertibility,

$$\phi(L)x_t = \varepsilon_t, \quad \phi(L) = \frac{a(L)}{\beta(L)}.$$

This AR(∞) representation expresses the innovation using current and past observations. For a causal, invertible ARMA process, the MA representation is its Wold innovation representation. More generally, Wold's theorem also concerns the nondeterministic component of a covariance-stationary process.

Question 31: ARMA(1,1) expansion and cancellation

For

$$x_t = ax_{t-1} + \varepsilon_t + b\varepsilon_{t-1},$$

derive the MA(∞) coefficients. What happens if $b = -a$?

Worked solution

If $|a| < 1$,

$$\theta(L) = \frac{1 + bL}{1 - aL}.$$

Use the geometric expansion

$$\frac{1}{1 - aL} = 1 + aL + a^2 L^2 + \dots$$

Multiplying,

$$\theta(L) = 1 + (a + b)L + a(a + b)L^2 + a^2(a + b)L^3 + \dots$$

Therefore,

$$\boxed{\theta_0 = 1, \quad \theta_j = a^{j-1}(a + b) \quad (j \geq 1).}$$

If $b = -a$, then

$$\frac{1 - aL}{1 - aL} = 1,$$

so $x_t = \varepsilon_t$ in the stationary solution.

The written ARMA(1,1) has redundant common factors and reduces to white noise. Identification rules should be applied to the minimal representation.

5 Forecasting and Forecast Uncertainty

Question 32: Why the conditional mean is the optimal forecast

Show that the conditional expectation minimizes expected squared forecast error.

Worked solution

Let

$$m = \mathbb{E}(y_{T+h} \mid \mathcal{F}_T)$$

and let g be any forecast measurable using $\mathcal{F}_T$. Write

$$y_{T+h} - g = (y_{T+h} - m) + (m - g).$$

Taking conditional expectations of squared errors,

$$\mathbb{E}[(y_{T+h} - g)^2 \mid \mathcal{F}_T] = \mathbb{E}[(y_{T+h} - m)^2 \mid \mathcal{F}_T] + (m - g)^2,$$

because

$$\mathbb{E}(y_{T+h} - m \mid \mathcal{F}_T) = 0.$$

The first term does not depend on g . The second is minimized at $g = m$.

Thus

$$\hat{y}_{T+h|T} = \mathbb{E}(y_{T+h} \mid \mathcal{F}_T).$$

With only zero autocorrelation assumptions, linear projection methods deliver the best linear forecast. Equating that forecast to the full conditional mean needs additional assumptions.

Question 33: General recursive forecasting and a numerical example

Explain recursive ARMA forecasting. Then forecast one, two, and three periods ahead from

$$y_t = 1 + 0.5y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1},$$

given $y_T = 4$ and estimated $\varepsilon_T = 0.5$.

Worked solution

For each horizon:

- Use observed past outcomes when available.
- Replace future outcomes on the right-hand side with earlier forecasts.
- Use estimated past innovations.
- Replace future innovations with their conditional expectation, zero.

The one-step forecast is

$$\hat{y}_{T+1|T} = 1 + 0.5(4) + 0.4(0.5) = 3.2.$$

At two steps, ε_{T+1} has conditional mean zero:

$$\hat{y}_{T+2|T} = 1 + 0.5(3.2) = 2.6.$$

Similarly,

$$\hat{y}_{T+3|T} = 1 + 0.5(2.6) = 2.3.$$

The mean is

$$\mu = \frac{1}{1 - 0.5} = 2,$$

and forecasts converge to 2.

Past MA innovations cease to enter directly once the forecast horizon exceeds q , though their influence can continue indirectly through the AR recursion.

Question 34: Forecast-error variance from the MA representation

Derive the h -step forecast-error variance. Distinguish innovation uncertainty from estimation uncertainty.

Worked solution

Suppose

$$y_t - \mu = \sum_{j=0}^{\infty} \theta_j \varepsilon_{t-j}, \quad \theta_0 = 1.$$

With known parameters and known past innovations, the unpredictable part of y_{T+h} is

$$e_{T+h|T} = \sum_{j=0}^{h-1} \theta_j \varepsilon_{T+h-j}.$$

Since innovations are uncorrelated,

$$\text{Var}(e_{T+h|T}) = \sigma_{\varepsilon}^2 \sum_{j=0}^{h-1} \theta_j^2.$$

The variance is nondecreasing in h because

$$\text{Var}(e_{T+h+1|T}) - \text{Var}(e_{T+h|T}) = \sigma_{\varepsilon}^2 \theta_h^2 \geq 0.$$

This formula captures future-innovation uncertainty. Using estimated parameters adds parameter uncertainty. A finite sample can also introduce uncertainty from initialization or imperfectly recovered past innovations.

Under conditional homoskedasticity, the same expression is the conditional forecast-error variance.

Question 35: AR(1) forecast variance and prediction intervals

Suppose

$$y_t = 2 + 0.8y_{t-1} + \varepsilon_t, \quad \sigma_\varepsilon^2 = 4, \quad y_T = 12.$$

Find the one-step and three-step forecasts, their forecast-error variances, and the long-run limits.

Worked solution

The stationary mean is

$$\mu = \frac{2}{1 - 0.8} = 10.$$

Thus

$$\hat{y}_{T+h|T} = 10 + 0.8^h(12 - 10).$$

In particular,

$$\hat{y}_{T+1|T} = 11.6, \quad \hat{y}_{T+3|T} = 11.024.$$

For AR(1), $\theta_j = 0.8^j$, so

$$V_h = 4 \sum_{j=0}^{h-1} 0.8^{2j} = 4 \frac{1 - 0.8^{2h}}{1 - 0.8^2}.$$

Therefore,

$$V_1 = 4,$$

$$V_3 = 4(1 + 0.64 + 0.4096) = 8.1984.$$

As $h \rightarrow \infty$,

$$\hat{y}_{T+h|T} \rightarrow 10, \quad V_h \rightarrow \frac{4}{1 - 0.64} = 11.1111.$$

If innovations are Gaussian and parameters are treated as known, a 95% prediction interval is

$$\hat{y}_{T+h|T} \pm 1.96\sqrt{V_h}.$$

For one step,

$$11.6 \pm 1.96(2) = [7.68, 15.52].$$

This predicts a future observation; it is not merely a confidence interval for its conditional mean.

6 Impulse Response Functions

Question 36: Meaning, scaling, and interpretation of an IRF

Define an impulse response and relate it to MA coefficients. Explain why the intercept cancels and distinguish an innovation response from a causal policy effect.

Worked solution

The response at horizon h to a unit innovation at time t is

$$\text{IRF}(h) = \frac{\partial y_{t+h}}{\partial \varepsilon_t}.$$

For the linear representation

$$y_{t+h} - \mu = \sum_{j=0}^{\infty} \theta_j \varepsilon_{t+h-j},$$

the coefficient on ε_t is θ_h , so

$$\boxed{\text{IRF}(h) = \theta_h.}$$

Equivalently, compare two expected paths with the same history: one has $\varepsilon_t = 1$ and the other has $\varepsilon_t = 0$. Their difference at horizon h is θ_h .

The intercept is identical in both paths and cancels. A shock of size c produces response $c\theta_h$. A one-standard-deviation shock produces $\sigma_\varepsilon \theta_h$.

An ARMA innovation is an unanticipated statistical disturbance. Calling its response the causal effect of a specific policy requires additional structural identification.

Question 37: Two methods for computing IRFs and their uncertainty

Describe the recursive-path and polynomial-expansion methods. Why does IRF uncertainty differ from forecast uncertainty?

Worked solution

The recursive-path method computes:

- (a) A baseline expected path with the time- t shock set to zero.
- (b) A second expected path with that shock set to one.
- (c) Their difference at every horizon.

All other shocks and initial conditions are identical.

The polynomial method expands

$$\theta(L) = \frac{\beta(L)}{a(L)} = 1 + \theta_1 L + \theta_2 L^2 + \dots$$

and reads $\text{IRF}(h) = \theta_h$.

For a fixed linear model, known parameters imply a known IRF. Parameter estimates create uncertainty about the response coefficients.

A future realized outcome remains uncertain even with known parameters because future innovations are unknown. Thus the forecast-error variance formula is not an IRF confidence-band formula.

Question 38: Lecture ARMA(2,1): undetermined coefficients

For the lecture model

$$y_t = 1.2y_{t-1} - 0.7y_{t-2} + \varepsilon_t + 0.3\varepsilon_{t-1},$$

derive the first six IRF coefficients, describe their pattern, and compute the cumulative response.

Worked solution

The polynomials are

$$a(L) = 1 - 1.2L + 0.7L^2, \quad \beta(L) = 1 + 0.3L.$$

Set

$$\frac{1 + 0.3L}{1 - 1.2L + 0.7L^2} = 1 + \theta_1 L + \theta_2 L^2 + \dots$$

Multiplying both sides by the denominator gives

$$1 + 0.3L = (1 - 1.2L + 0.7L^2)(1 + \theta_1 L + \theta_2 L^2 + \dots).$$

Equating coefficients:

$$\theta_0 = 1,$$

$$\theta_1 - 1.2 = 0.3 \Rightarrow \theta_1 = 1.5,$$

$$\theta_2 - 1.2\theta_1 + 0.7 = 0 \Rightarrow \theta_2 = 1.1.$$

For $h \geq 2$,

$$\boxed{\theta_h = 1.2\theta_{h-1} - 0.7\theta_{h-2}.}$$

Hence

$$\theta_3 = 1.2(1.1) - 0.7(1.5) = 0.27,$$

$$\theta_4 = 1.2(0.27) - 0.7(1.1) = -0.446,$$

$$\theta_5 = 1.2(-0.446) - 0.7(0.27) = -0.7242.$$

The dynamic roots solve

$$r^2 - 1.2r + 0.7 = 0$$

and equal

$$r = 0.6 \pm i\sqrt{0.34}.$$

Their modulus is $\sqrt{0.7} < 1$, so responses show damped oscillation and eventually converge to zero.

Because the coefficients are absolutely summable,

$$\sum_{h=0}^{\infty} \theta_h = \theta(1) = \frac{1 + 0.3}{1 - 1.2 + 0.7} = \boxed{2.6}.$$

This is the cumulative response over time, not the limiting response at an individual distant horizon, which is zero.

7 Integrated Review

Question 39: Complete empirical workflow

Explain how to move from a raw macroeconomic or financial series to a defensible ARMA model and then to forecasts and impulse responses.

Worked solution

A coherent workflow is:

- (a) Plot the raw series and inspect trend, seasonality, unusual observations, and possible changes in behavior.
- (b) Consider transformations, such as logs and first differences, when stationarity is doubtful.
- (c) Examine the transformed series' SACF and SPACF to propose a small set of candidate orders.
- (d) Estimate pure AR models by OLS or likelihood and models with MA terms by NLLS or likelihood.
- (e) Check stationarity, invertibility, and the meaning of the reported constant.
- (f) Compare compatible AIC/BIC values using the same sample.
- (g) Examine residual correlograms and residual Ljung–Box tests.
- (h) Revise inadequate models rather than selecting mechanically from an information criterion.
- (i) Produce recursive forecasts and quantify uncertainty.
- (j) Derive MA coefficients to interpret shock propagation and assess parameter uncertainty in IRFs.

Predictive adequacy and causal interpretation remain separate questions throughout.

Question 40: Resolving conflicting selection evidence

AR(1) has lower BIC but a residual Ljung–Box p -value of 0.01. ARMA(1,1) has slightly higher BIC and a residual p -value of 0.48. Which model is more defensible? Does 0.48 prove correctness?

Worked solution

The AR(1) leaves statistically detectable autocorrelation, suggesting that it has not captured the conditional-mean dynamics adequately.

The ARMA(1,1) is more defensible on the evidence given, provided its parameters are admissible and its other diagnostics are satisfactory. A small BIC improvement does not eliminate the need for residual adequacy.

However, $p = 0.48$ does not prove that ARMA(1,1) is correct. The test has limited power, examines selected lags, and does not test every form of misspecification.

The practical conclusion is that ARMA(1,1) survives this diagnostic while AR(1) does not, rather than that either model has been established as the true data-generating process.

Coverage Checklist

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Compact Formula Sheet

Classical regression

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad \text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

$$\text{MSE}(\hat{\beta}) = \text{Var}(\hat{\beta}) + \text{Bias}(\hat{\beta})^2.$$

Stationary time series

$$\mathbb{E}(y_t) = \mu, \quad \gamma_k = \text{Cov}(y_t, y_{t-k}), \quad \rho_k = \frac{\gamma_k}{\gamma_0}.$$

$$\phi_{11} = \rho_1, \quad \phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

AR(1)

$$\mu = \frac{a_0}{1 - a_1}, \quad \gamma_0 = \frac{\sigma_\varepsilon^2}{1 - a_1^2}, \quad \rho_k = a_1^k.$$

$$\hat{y}_{T+h|T} = \mu + a_1^h(y_T - \mu).$$

$$V_h = \sigma_\varepsilon^2 \frac{1 - a_1^{2h}}{1 - a_1^2}.$$

Model comparison and diagnostics

$$AIC = -2\hat{\ell} + 2k, \quad BIC = -2\hat{\ell} + k \log T.$$

$$Q_K = T(T+2) \sum_{j=1}^K \frac{r_j^2}{T-j}.$$

For fitted ARMA(p, q) residuals, the lecture uses

$$Q_K \stackrel{a}{\sim} \chi_{K-p-q}^2.$$

ARMA representations and inference

For $x_t = y_t - \mu$,

$$a(\mathbf{L})x_t = \beta(\mathbf{L})\varepsilon_t,$$

$$\theta(\mathbf{L}) = \frac{\beta(\mathbf{L})}{a(\mathbf{L})}, \quad \phi(\mathbf{L}) = \frac{a(\mathbf{L})}{\beta(\mathbf{L})}.$$

$$\text{Var}(y_{T+h} - \hat{y}_{T+h|T}) = \sigma_\varepsilon^2 \sum_{j=0}^{h-1} \theta_j^2, \quad \text{IRF}(h) = \theta_h.$$